

ORIGINAL ARTICLE

Discriminant analysis of game-related statistics between basketball guards, forwards and centres in three professional leagues

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Abstract

The aim of the present paper was to examine the differences in game-related statistics between basketball guards, forwards and centres playing in three professional leagues: National Basketball Association (NBA, superior level) in the USA, Asociación de Clubs de Baloncesto (ACB, one of the best European leagues) in Spain and Liga de Clubes de Basquetebol (LCB, inferior level) in Portugal. We reasoned that the knowledge of these differences could allow the coaches to better establish and monitor playing patterns and increase the effectiveness of the player recruitment process. Archival data was gathered for the 2000–2001 play-off final series of the NBA (five games), ACB (three games) and LCB (four games). For players in each league, discriminant analysis was able to identify game-related statistics that maximized mean differences between playing positions ($p < 0.05$). The interpretation of the obtained discriminant functions was based on examination of the structure coefficients greater than $|0.30|$. In the LCB league, centres and guards were discriminated mainly in terms of defensive tasks, with emphasis on blocks (structure coefficient, $SC = 0.35$) and defensive rebounds ($SC = 0.43$) and a de-emphasis on unsuccessful 3-point field-goals ($SC = -0.37$). In the ACB, centres and guards were discriminated by offensive tasks, with emphasis on assists ($SC = 0.52$) and 3-point field-goals, both successful ($SC = 0.35$) and unsuccessful ($SC = 0.35$), and a de-emphasis on offensive rebounds ($SC = -0.44$). Finally, in the NBA league guards and centres were discriminated by offensive tasks, with emphasis on offensive rebounds ($SC = 0.31$) and a de-emphasis on assists ($SC = -0.37$) and unsuccessful 3-point field-goals ($SC = -0.34$). These three analyses provided high overall percentages of successful classification (86% for the LCB league, 74% for the ACB and 85% for the NBA). Generally, the players' game-related statistics varied according to playing position, probably because of the well-known differences in the players' anthropometric characteristics that conditioned the distance they play from the basket. Coaches can use these results to reinforce the importance of relying on different players' contributions to team performance and evaluate players' game performance according to their playing position. Conversely, these discriminant models could help in player recruitment and improve training programmes.

Keywords: *Basketball, discriminant analysis, player position, professional leagues, game statistics*

Introduction

At the highest level of basketball competition, success can be mostly dependent on the recruitment of players with complementary skills who are capable of performing according to the demands of their playing position (Ige & Kleiner, 1998). Over the past few years, the contrast between basketball guards, forwards and centres has been addressed only through the measurement and evaluation of somatotype, body composition and physiological responses. Results from these studies concerning

female basketball players have shown differences between anthropometric and physiological measures of individuals in these playing positions (Ackland, Schreiner, & Kerr, 1997; Bale, 1991; Smith & Thomas, 1991). For example, it was showed that centres were much taller, had longer limb lengths, longer hip widths and were more muscular than guards and forwards (Bale, 1991).

A very important area of research that has not been adequately investigated is the relationship between a player's position on a team and their

game-related statistics profile. In fact, quantitative analysis of basketball performance through game-related statistics is being widely used amongst coaches in order to analyse game events with more valid data. However, available research on this subject has only addressed the identification of the most discriminant statistics between winning and losing teams (e.g., Akers, Wolff, & Buttross, 1991; Ibáñez, Sampaio, Sáenz-López, Giménez, & Janeira, 2003; Sampaio & Janeira, 2003; Trninić, Milanović, & Dizdar, 1997). This would seem to be an important area of research, because demands placed upon players differ as a function of playing position. For example, it would seem logical that taller and stronger players are selected as centres in order to play nearer to the basket and secure more rebounds. Therefore, it seems reasonable to suspect that game-related statistical performance is linked to playing position.

On the other hand, the standard of competition can also affect players' game-related statistical profiles, i.e., characteristics of different leagues might have an impact upon players' game-related statistical profiles. However, these are only perceptions; we could find no studies addressing these particular questions. The knowledge of these results could allow the coaches to establish and monitor playing patterns and increase the effectiveness of the player recruitment process. In fact, the evaluation of players' game performance must be done according to specific normative data regarding their playing position. Additionally, it can allow coaches to understand better the differences between leagues and have a more precise idea on how some players could perform in another league.

Thus, the purpose of this investigation was to study the discriminant power of game-related statistics between players' position (guard, forward and centre) at three different standards of male basketball competition: National Basketball Association (NBA) in the USA, Asociación de Clubs de Baloncesto (ACB) in Spain and Liga de Clubes de Basquetebol (LCB) in Portugal.

Methods

Subjects

Archival data were obtained from official box scores for the 2000–2001 play-off final series of the NBA (five games), ACB (three games) and LCB (four games). The game-related statistics gathered included: 2- and 3-point field-goals (both successful and unsuccessful), free-throws (both successful and unsuccessful), defensive and offensive rebounds, blocks, assists, fouls, steals, turnovers and minutes

played. All data were gathered by each leagues' professional technicians.

As previously suggested the players were subdivided so that the "point guards" and "offguards" were pooled as guards ($n=93$), the "small forwards" and "power forwards" were grouped as forwards ($n=95$) and the centres ($n=64$) formed the third group for analysis (Ackland et al., 1997; Spurgeon, Spurgeon, & Giese, 1981). Players whose participation in any game was for less than 5 minutes' duration were excluded from the analysis.

Data analysis

In order to compare the game-related statistics collected between the players in each of the three leagues, each player's results were divided by that player's duration on court, resulting in derived-rate variables.

Subsequently, discriminant analysis was performed in order to determine: i) which of the obtained variables are more useful in predicting player position; ii) the mathematical equation that enhanced differences in variable means between guards, forwards and centres, and, iii) the accuracy of the equations. Assumptions on discriminant analysis were for independency amongst variables, multivariate normal distribution and equal variance-covariance across groups (Silva & Stam, 1995).

The variables in our study are derived-rate variables (because the original game-related statistics were divided by the time played by each player), the discriminant analysis is considered to be robust with these variables (Norušis, 1993). The interpretation of the obtained discriminant functions was based on examination of the structure coefficients greater than $|0.30|$, meaning that variables with higher absolute values have a powerful contribution to discriminate between groups (Tabachnick & Fidell, 2000). Validation of discriminant models was conducted using the leave-one-out method of cross-validation (Norušis, 1993). Cross-validation analysis takes subsets of data for training and testing and is needed in order to understand the usefulness of discriminant functions when classifying new data. This method involves generating the discriminant function on all but one of the participants ($n-1$) and then testing for group membership on that participant. The process is repeated for each participant (n times) and the percentage of correct classifications generated through averaging for the n trials. The statistical analyses were performed using SPSS software release 10.0.1 and significance was set at $p \leq 0.05$.

Table I. Descriptive results from the game-related statistics for the three leagues (values are mean $\pm$ S.D. counts per minutes played).

Variable	LCB			ACB			NBA		
	Guards (<i>n</i> = 14)	Forwards (<i>n</i> = 28)	Centres (<i>n</i> = 22)	Guards (<i>n</i> = 18)	Forwards (<i>n</i> = 24)	Centres (<i>n</i> = 15)	Guards (<i>n</i> = 43)	Forwards (<i>n</i> = 28)	Centres (<i>n</i> = 17)
Assists	4.0 $\pm$ 3.5	1.2 $\pm$ 1.1	1.7 $\pm$ 1.7	2.2 $\pm$ 1.6	1.3 $\pm$ 1.3	0.6 $\pm$ 0.8	3.4 $\pm$ 2.7	1.1 $\pm$ 1.6	1.7 $\pm$ 2.5
Blocks	0.0 $\pm$ 0.0	0.2 $\pm$ 0.4	1.0 $\pm$ 1.1	0.2 $\pm$ 0.5	0.6 $\pm$ 1.2	0.5 $\pm$ 0.6	0.4 $\pm$ 0.7	0.8 $\pm$ 0.9	1.7 $\pm$ 2.4
Defensive rebounds	1.7 $\pm$ 1.5	1.6 $\pm$ 1.6	4.8 $\pm$ 2.3	1.8 $\pm$ 1.6	2.2 $\pm$ 2.3	2.1 $\pm$ 1.2	2.8 $\pm$ 2.6	3.2 $\pm$ 2.1	5.5 $\pm$ 4.7
Fouls	2.1 $\pm$ 1.7	1.9 $\pm$ 1.4	3.0 $\pm$ 1.3	2.9 $\pm$ 1.4	2.3 $\pm$ 1.3	3.3 $\pm$ 1.1	2.4 $\pm$ 1.4	2.5 $\pm$ 1.5	3.9 $\pm$ 1.8
Offensive rebounds	0.7 $\pm$ 0.8	0.8 $\pm$ 0.8	1.6 $\pm$ 1.3	0.6 $\pm$ 0.7	1.2 $\pm$ 1.1	1.7 $\pm$ 1.3	0.8 $\pm$ 1.3	1.4 $\pm$ 1.2	3.3 $\pm$ 2.7
Steals	1.9 $\pm$ 2.1	1.1 $\pm$ 1.0	1.0 $\pm$ 1.0	1.1 $\pm$ 1.3	1.2 $\pm$ 1.1	0.8 $\pm$ 1.1	1.4 $\pm$ 1.2	0.5 $\pm$ 0.8	0.3 $\pm$ 0.5
Successful 2-point field-goals	2.1 $\pm$ 2.0	1.7 $\pm$ 1.9	3.7 $\pm$ 2.4	1.4 $\pm$ 1.4	2.5 $\pm$ 1.6	2.1 $\pm$ 2.1	3.2 $\pm$ 3.8	1.6 $\pm$ 1.7	7.0 $\pm$ 5.6
Successful 3-point field-goals	1.1 $\pm$ 1.0	1.1 $\pm$ 1.1	0.4 $\pm$ 0.7	1.0 $\pm$ 1.1	0.9 $\pm$ 1.0	0.2 $\pm$ 0.5	0.9 $\pm$ 1.2	0.6 $\pm$ 1.1	0.0 $\pm$ 0.0
Successful free-throws	1.9 $\pm$ 2.1	1.0 $\pm$ 1.5	2.7 $\pm$ 2.6	2.4 $\pm$ 2.0	2.1 $\pm$ 2.2	2.0 $\pm$ 1.7	2.5 $\pm$ 3.2	1.1 $\pm$ 1.7	3.5 $\pm$ 3.5
Turnovers	2.6 $\pm$ 2.1	1.3 $\pm$ 1.3	2.2 $\pm$ 1.3	1.6 $\pm$ 1.4	1.7 $\pm$ 1.0	1.1 $\pm$ 0.7	1.9 $\pm$ 1.4	0.8 $\pm$ 1.3	1.7 $\pm$ 1.7
Unsuccessful 2-point field-goals	2.1 $\pm$ 1.4	1.1 $\pm$ 1.1	3.7 $\pm$ 2.2	2.0 $\pm$ 1.5	2.6 $\pm$ 1.8	1.1 $\pm$ 1.1	4.8 $\pm$ 4.9	2.6 $\pm$ 2.2	5.1 $\pm$ 4.0
Unsuccessful 3-point field-goals	1.9 $\pm$ 1.3	2.6 $\pm$ 1.7	0.9 $\pm$ 1.6	1.8 $\pm$ 1.4	1.3 $\pm$ 1.5	0.6 $\pm$ 0.9	1.5 $\pm$ 1.8	0.6 $\pm$ 0.9	0.0 $\pm$ 0.0
Unsuccessful free-throws	0.6 $\pm$ 0.9	0.4 $\pm$ 0.6	1.1 $\pm$ 1.0	0.8 $\pm$ 1.4	0.7 $\pm$ 1.0	1.1 $\pm$ 1.2	0.7 $\pm$ 1.8	0.2 $\pm$ 0.5	2.7 $\pm$ 3.9
Duration on court (min)	24.7 $\pm$ 13.0	21.5 $\pm$ 9.0	29.9 $\pm$ 9.6	19.2 $\pm$ 7.5	22.4 $\pm$ 7.4	18.0 $\pm$ 5.7	29.6 $\pm$ 14.4	22.2 $\pm$ 10.4	30.1 $\pm$ 16.7

Key: NBA, National Basketball Association in the USA; ACB, Asociación de Clubs de Baloncesto in Spain; LCB, Liga Portuguesa de Basquetebol in Portugal.

Results

The means and standard deviations for each group of basketball players for the studied game-related statistics are presented in Table I.

As shown in Table II and Figure 1, the group centroid distances (especially for the first discriminant function) and structure coefficients, describe the game-related statistical profiles that differentiate between guards, forwards and centres of the leagues

studied. The structure coefficients quantify the potential of each game-related statistic to maximize differences between means amongst guards, forwards and centres. The larger the magnitude of the coefficients, the greater the contribution of that game-related statistic to the discriminant function.

In the LCB league, discriminant function 1 accounted for 70% of the variance, whilst discriminant function 2 accounted for the remaining 29.7%. Results from function 1 reflect an emphasis on

Table II. Discriminant function structure coefficients and tests of statistical significance.

Variable	LCB		ACB		NBA	
	Function 1	Function 2	Function 1	Function 2	Function 1	Function 2
Assists	-0.19	0.52	0.47	0.19	-0.37	0.40
Blocks	0.35	-0.09	-0.23	-0.19	0.25	-0.13
Defensive rebounds	0.43	0.13	-0.05	0.19	0.22	-0.32
Fouls	0.14	-0.01	-0.18	0.62	0.27	0.09
Offensive rebounds	0.14	-0.16	-0.44	-0.10	0.31	-0.15
Steals	-0.16	0.03	-0.06	-0.08	-0.24	0.04
Successful 2-point field-goals	0.24	-0.01	-0.20	-0.37	0.22	0.34
Successful 3-point field-goals	-0.26	-0.21	0.35	-0.13	-0.20	-0.14
Successful free-throws	0.10	0.16	0.08	0.32	0.02	0.43
Turnovers	-0.09	0.37	0.20	0.02	-0.02	0.29
Unsuccessful 2-point field-goals	0.24	0.21	0.18	-0.44	0.15	0.19
Unsuccessful 3-point field-goals	-0.37	-0.45	0.35	0.21	-0.34	0.02
Unsuccessful free-throws	0.11	0.17	-0.10	0.25	0.13	0.41
Wilks' Lambda	0.182	0.543	0.363	0.678	0.219	0.636
Chi-square	93.8	33.5	48.6	18.6	120.0	35.7
<i>P</i>	<0.001	<0.001	<0.05	N.S.	<0.001	<0.001
Eigenvalue	1.99	0.84	0.87	0.47	1.91	0.57
Relative percentage	70.33	29.67	64.63	35.37	76.97	23.03
Canonical correlation	0.82	0.68	0.68	0.57	0.81	0.60

Key: NBA, National Basketball Association in the USA; ACB, Asociación de Clubs de Baloncesto in Spain; LCB, Liga Portuguesa de Basquetebol in Portugal.

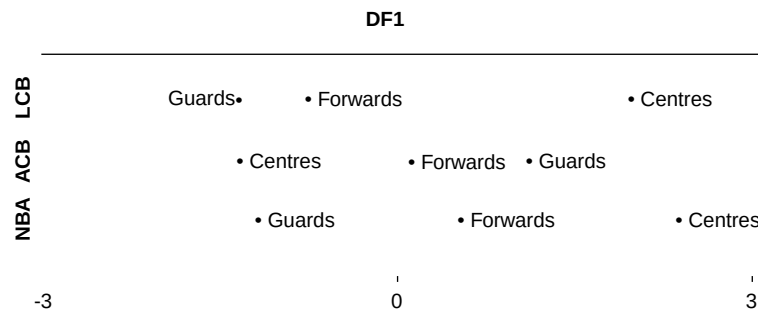


Figure 1. Territorial map of the players relative to their playing position representing how widely dispersed the centroids are from one another in standardised discriminant scores. The points indicate the group centroid for guards, forwards and centres. DF1 = discriminant function 1.

blocks and defensive rebounds and a de-emphasis on unsuccessful 3-point field-goals (see Table II).

On the other hand, in the ACB league, discriminant function 1 accounted for 64.6% of the variance. The remaining variance was accounted for by discriminant function 2. However, this function failed to reach statistical significance. In this league, the structure coefficients from function 1 reflect an emphasis on assists and 3-point field-goals (both successful and unsuccessful) and a de-emphasis on offensive rebounds (see Table II).

Finally, in the NBA league, discriminant function 1 accounted for 77.0% of the variance, whilst discriminant function 2 accounted for the remaining 23.0%. The structure coefficients from function 1 reflect an emphasis on offensive rebounds and a de-emphasis on assists and unsuccessful 3-point field-goals (see Table II).

The leave-one-out test summarises the ability of the discriminant functions to correctly classify the players in their respective positions (see Table III). This analysis provided an overall percentage of successful classification of 85.9% for the LCB league, 73.7% for the ACB and 85.2% for the

NBA. Notably, almost all LCB centres and NBA guards were correctly classified on the basis of their game-related statistics.

Discussion

The purpose of this investigation was to study the discriminant power of basketball game-related statistics between players' position (guard, forward and centre) at three different standards of competition: the NBA in the USA, the ACB league in Spain and the LCB league in Portugal. The game-related statistics for a sample of players from the final series of the 2000–2001 play-offs were analysed specifically to characterize the players from the best teams and their performances in the most critical games. In this way, it is more likely that the best players get more time on court and that their performances represent best their real differences (Sampaio & Janeira, 2003). Additionally, this is the time of the season in which the players will be in their best physical and psychological condition.

The results of this study are new insights to the understanding of the basketball game. In fact, our results describe precisely the actions that distinguish players by their position and allow us to better understand how team performance depends upon players with complementary skills. These profiles are in the origin of player physiological demands and can be helpful to plan specific training programmes. For example, the exercises that are specific to the energy and muscle demands of the basketball game can now be also specific to the technical actions.

These player performance profiles also varied across the leagues studied denoting that standard of play is an important factor to take into account. However, all of them seem much related to the distance that players are from the basket. Therefore, it seems that players' anthropometric status could also have influenced these results, because some body sizes are more suitable to the demands of some playing positions (Ackland et al., 1997), i.e., it is more logical to select the taller players as centres.

Table III. Classification matrix for the players' actual and predicted playing position according to game-related-statistics of the discriminant functions.

	Actual group	Predicted group		
		Guards	Forwards	Centres
LCB	Guards ($n=14$)	79%	14%	7%
	Forwards ($n=28$)	11%	82%	7%
	Centres ($n=22$)	0%	5%	96%
ACB	Guards ($n=18$)	78%	11%	11%
	Forwards ($n=24$)	17%	75%	8%
	Centres ($n=15$)	7%	27%	67%
NBA	Guards ($n=43$)	98%	3%	0%
	Forwards ($n=28$)	25%	75%	0%
	Centres ($n=17$)	0%	29%	71%

Key: NBA, National Basketball Association in the USA; ACB, Asociación de Clubs de Baloncesto in Spain; LCB, Liga Portuguesa de Basquetebol in Portugal.

The ability of the discriminant functions in correctly classifying the players in their respective positions was high, denoting the quality of the discriminant functions and the power of the structure coefficients in explaining variability amongst groups.

In the LCB league, most of this variability (70%, first discriminant function) is accounted to describe differences between centres and guards mainly in defensive tasks. In fact, centres are required to utilise their size (height and body mass) to the benefits of the team in defensive rebounding and blocking. Additionally, centres also missed less 3-point field-goals than the guards and forwards. This is the result of a lower number of attempts because these players are near to the basket most of the time and consequently are more specialised in attempting inside field-goals (see Table I).

The remaining variability (30%, second discriminant function) seems accounted to describe differences between guards and forwards in terms of offensive tasks. The guards tend to effect quick transitions from defensive to attacking patterns and control the flow of these patterns. This requires them to master ball-handling skills such as ball control, dribble penetration and passing, which might explain the discriminatory power of assists and turnovers. Once again, the unsuccessful 3-point field-goals were able to discriminate player positions, this probably occurred because the forwards tend to be specialists in long-distance shooting so they have a higher number of attempts and consequently higher number of misses. On the other hand, the guards' tasks include preparing offensive situations so that the forwards can have better opportunities to shoot (e.g., with a dribble penetration the guard can force a defensive help from the weak side of the offence and then assist to the open forward). Considering that the guards are the players subjected to a higher level of defensive pressure (Trninić et al., 1997), these tasks do not leave them much time and/or space to increase the frequency of outside shoots.

In the ACB league, 65% of the variability amongst groups (first discriminant function) seems accounted to describe differences between centres and guards in offensive tasks. As seen in Figure 1, the differences between player position centroids were the narrowest, which allows us to consider that the players' game-related statistical profiles were more homogeneous. The fact that the second discriminant function failed to reach statistical significance might confirm this idea. In this league, the guards' statistical profile was characterized by strong passing skills and 3-point field-goals (both successful and unsuccessful). These last two variables seem to attribute them a more important role in outside shooting. On

the other hand, the de-emphasis on offensive rebounds might be explained by basket proximity.

In the NBA league, most of the variability amongst groups (77%, first discriminant function) was accounted to describe differences between guards and centres in offensive tasks. We have identified a larger distance between player positions centroids denoting stronger differences between players' game-related statistical profiles. The explanation of this fact may be in the centres' anthropometric characteristics and in teams' style of play (which certainly influences player recruitment). In fact, NBA centres seem to be larger (height and mass) than the other players, and these characteristics suggest that they are highly specialized in rebounding, inside shooting, screening or drawing fouls and 3-point plays. Additionally, it suggests that their participation in the game is confined to near the basket. Confirmation of this idea is strengthened by fact that the players from the analyzed sample did not attempt even one 3-point field-goal, probably because the 3-point line is farther from the basket and playing farther from the basket implies leaving other roles like rebounding. Additionally, the second discriminant function (23% of the variability) can also help us to better understand the results because it describes differences between centres and forwards in offensive and defensive tasks to a larger extent (in assists, defensive rebounds, successful 2-point field-goals and free throws—both successful and unsuccessful) creating a profile that seems to suite an all-round player. In this way, our results seem to suggest that NBA guards and centres are highly specialized players whereas the forwards role is much more flexible.

In practical applications, the players' game-related statistical profiles varied according to playing position and across the studied leagues. Coaches can use these results to make training programmes more specific, e.g., guards from each league should spend more time improving their efficacy in tasks related to their specific game-related statistics profile (identified in our results). The result of this work reinforces the importance of relying on different players' contributions to team performance.

High-level selectors can use these results to: i) select basketball players according to complementary specific profiles and; ii) have a more precise assessment of the impact of changing to another league upon a players' game-related statistical profile, e.g., selecting guards from LCB with very few number of turnovers or, on the other hand, selecting centres from LCB and ACB with a very high number of offensive rebounds. Thus, it seems appropriate to evaluate players' game performance according to normative data of their playing position. The territorial map (see Figure 1) provided an initial model for discriminating basketball players. In this manner,

players' performances evaluated in other leagues could be compared with group centroids. Additionally, the territorial map gives us the new information that in LCB the centres exhibit a clearly distanced performance profile (probably denoting in these leagues, the importance of players' height and weight). However, ACB players' game-related statistical profiles were closer to each other, giving us the idea that players could be less specialised in their roles. Finally, in the NBA the differences between positions seem to be wide, but here we think that this fact can be much attributed to rule differences (e.g., the 3-point line is one meter more distant from the basket). These results also seem to suggest that top-level players' game-related statistical profiles could be more homogeneous because players seem to be less specialized in their roles (they are all tall, they are all good shooters and passers, ...).

Keypoints

- The players' game-related statistics varied according to playing position and between the leagues mainly as a consequence from the distance they play from the basket.
- The result of this work reinforces the importance of relying on different players' contributions to team performance.
- In LCB the centres exhibit a clearly distanced performance profile (probably denoting in these leagues, the importance of players' height and weight). The ACB players' game-related statistical profiles were closer to each other, giving us the idea that players could be less specialised in their roles. In the NBA the differences between positions seem to be wide, but here we think that this fact can be much attributed to rule

differences (e.g., the 3-point line is one meter more distant from the basket).

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